

HDOC DAY-21

C Programming Activity

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Problem Statement

Write a menu-driven C program that accepts two positive integers **m** and **n**, where $m \leq n$. Use a **switch** statement and repeatedly display a menu to: (1) print the first 10 multiples of each integer from m to n, (2) print addition tables using addends 1 to 10, (3) print subtraction tables using subtrahends 1 to 10, (4) divide each integer by 1 to 10 and display results to two decimal places, and (5) exit. Display an error message for an invalid choice.

Algorithm

1. Start the program and read positive integers m and n.
2. If $m > n$ or either value is not positive, display an error and stop.
3. Display the menu inside a do-while loop and read the user's choice.
4. Use switch-case. For choices 1-4, use nested for loops for every number from m to n and values 1 to 10.
5. Choice 1: print number $\times$ i. Choice 2: print number $+$ i. Choice 3: print number $-$ i.
6. Choice 4: print number $/$ i using floating-point division with two decimal places.
7. Choice 5 exits. For any other choice, print an invalid-choice message.
8. Repeat until choice 5 is selected, then stop.

C Program

```
#include <stdio.h>

int main() {
    int m, n, choice, num, i;

    printf("Enter m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n) {
        printf("Invalid input!\n");
        return 0;
    }

    do {
        printf("\n----- MENU ----- \n");
        printf("1. Multiplication tables\n");
        printf("2. Addition tables\n");
        printf("3. Subtraction tables\n");
        printf("4. Division tables\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                for (num = m; num <= n; num++) {
                    printf("\nMultiples of %d:\n", num);
                    for (i = 1; i <= 10; i++)
                        printf("%d x %d = %d\n", num, i, num * i);
                }
                break;

            case 2:
                for (num = m; num <= n; num++) {
                    printf("\nAddition table of %d:\n", num);
                    for (i = 1; i <= 10; i++)
                        printf("%d + %d = %d\n", num, i, num + i);
                }
                break;

            case 3:
                for (num = m; num <= n; num++) {
                    printf("\nSubtraction table of %d:\n", num);
                    for (i = 1; i <= 10; i++)
                        printf("%d - %d = %d\n", num, i, num - i);
                }
                break;

            case 4:
                for (num = m; num <= n; num++) {
                    printf("\nDivision table of %d:\n", num);
                    for (i = 1; i <= 10; i++)
                        printf("%d / %d = %.2f\n", num, i,
                            (float)num / i);
                }
                break;

            case 5:
                printf("Program ended.\n");
                break;

            default:
                printf("Invalid menu choice!\n");
        }
    } while (choice != 5);

    return 0;
}
```

Test Case Table

Test	Input	Choice	Expected Result
1	m=2, n=3	1	First 10 multiples of 2 and 3
2	m=2, n=3	2	Addition tables of 2 and 3 (1 to 10)
3	m=2, n=3	3	Subtraction tables of 2 and 3 (1 to 10)
4	m=2, n=3	4	Division tables, rounded to 2 decimal places
5	m=2, n=3	7	Invalid menu choice!
6	m=2, n=3	5	Program ended
7	m=5, n=2	-	Invalid input!

Sample Output

```
Enter m and n (m <= n): 2 3
```

```
----- MENU -----
```

```
1. Multiplication tables
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit
```

```
Enter your choice: 4
```

```
Division table of 2:
```

```
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
...
2 / 10 = 0.20
```

```
Division table of 3:
```

```
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
...
3 / 10 = 0.30
```

```
Enter your choice: 5
```

```
Program ended.
```

Result: The menu-driven program was compiled and tested for the required arithmetic table operations.